

Lubna Zahan Lamia

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Research Assistant, Cognitive Agents and Interaction Lab (CAIL) — University of Dhaka, Bangladesh

RESEARCH PROFILE

Research Assistant at the Cognitive Agents and Interaction Lab (CAIL), University of Dhaka. Research interests lie at the intersection of **Natural Language Processing**, **Machine Learning**, and **human-AI alignment**, with a focus on how large language models construct, interpret, and frame human narratives. Current work includes semantic representation, sentiment modeling, and ethical AI through the analysis of perspective, bias, and framing in generative text, with an emphasis on developing alignment-focused frameworks for transparent, context-aware, and responsibly aligned AI behavior.

RESEARCH INTERESTS

- Natural Language Processing • Human-AI Alignment • Deep Learning • Bias and Perspective in NLP
- Ethical and Trustworthy AI

EDUCATION

University of Dhaka

Bachelor of Science in Computer Science & Engineering

- Cumulative GPA: 3.53 / 4.00
- Final Semester GPA: 3.89 / 4.00 · Ranked 3rd of 70

Dhaka, Bangladesh

Jan 2020 – July 2025

Holy Cross College

Higher Secondary Certificate (Science)

- Result: GPA 5.00 / 5.00; awarded Government Scholarship (H.S.C.).

Dhaka, Bangladesh

June 2017 – June 2019

Savar Cantonment Public School and College

Secondary School Certificate (Science)

- GPA: 5.00 / 5.00; awarded Government Scholarship (S.S.C.).

Dhaka, Bangladesh

Completed 2017

RESEARCH EXPERIENCE

Research Assistant

Cognitive Agents and Interaction Lab (CAIL), University of Dhaka

Jan 2024 – Present

Dhaka, Bangladesh

- Investigating human-AI collaboration dynamics in generative systems, with a focus on how humans and large language models co-construct meaning and interpretive outcomes.
- Conducting research on bias, framing, and interpretability in large language models.
- Developed analytical frameworks for narrative distortion and sentiment-informed modeling in generative systems.
- Contributed to peer-reviewed research in evaluation design, empirical analysis, and dataset development.
- Providing research mentorship to two undergraduate students in early-stage research development and experimental reasoning.

PUBLICATIONS AND PRESENTATIONS

- **Lamia, L. Z.**, Hossain, M. F. B., & Khan, M. M. (2025). *Who Holds the Pen? Caricature and Perspective in LLM Retellings of History*. *Conference on Empirical Methods in Natural Language Processing (EMNLP) 2025*, Suzhou, China.
Oral Presentation. Senior Area Chairs (SAC) Highlight — Outstanding Paper Nominee (track-specific).

- Hossain, M. F. B., Lamia, L. Z., & Khan, M. M. (2025). *FinBERT-BiLSTM: A Deep Learning Model for Predicting Volatile Cryptocurrency Market Prices Using Market Sentiment Dynamics*. Manuscript under final-stage review at *Applied Intelligence Journal (Springer)*; Equal contribution.

TECHNICAL SKILLS

Python, C++, Java | PyTorch, TensorFlow, Scikit-learn | Hugging Face (FinBERT, Sentence-BERT, RoBERTa-MNLI, LLaMA) | LLMs (model experimentation and evaluation using GPT, Claude, and Gemini) | Bi-LSTM, TCN, TFT, Informer | Git, Google Cloud

AWARDS & RECOGNITIONS

- **EMNLP 2025 Diversity & Inclusion Grant** recipient (USD 1000).
- **Full registration waiver** (USD 600) for EMNLP 2025 participation, including ACL membership.
- **Senior Area Chairs (SAC) Highlight Nominee** — track-specific Outstanding Paper nomination; among the 75 papers selected across all award categories from 8,174 EMNLP 2025 submissions.
- **World Youth Leadership Conference (WYLC) 2025**, Dubai — awarded partial scholarship from Global Entrepreneurship Bootcamp INC.
- **Government Scholarships** for academic merit in H.S.C., S.S.C., and J.S.C. examinations.

CO-CURRICULAR & LEADERSHIP ACTIVITIES

- Served as a **Student Volunteer** at the *Conference on Empirical Methods in Natural Language Processing (EMNLP) 2025*, Suzhou, China — In-person.
- **Departmental Coordination** — **Dept. of CSE, University of Dhaka** Led organization of *Graduation Week 2024*, coordinating multi-day events, scheduling, registration, and volunteer activities.
- **Program Host** — **University of Dhaka** Served as host for events including the *Graduation Ceremony*, *Freshers' Reception*, and *CAIL LLM Workshop*, supporting program flow and on-site coordination.

ACADEMIC REFERENCES

Dr. Md. Mosaddek Khan

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Principal Investigator, Cognitive Agents and Interaction Lab (CAIL)
Department of Computer Science and Engineering
University of Dhaka, Bangladesh
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Md. Mahmudur Rahman

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Mabsur Fatin Bin Hossain

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